

PakLaw AI

Pakistan Law AI Assistant — PRD

1. Goal

Build an AI assistant that answers users' questions using only the provided Pakistani legal documents. The system uses RAG so answers are grounded in retrieved legal text and show their sources.

2. Problem & Solution

Finding relevant sections/articles in lengthy Pakistani law documents is difficult. PakLaw AI lets a user ask a question in natural language, retrieves the most relevant legal chunks, and generates a concise answer with document/page/chunk references.

3. Current System Flow

Pakistan Law PDFs → PyMuPDF → Legal Chunks → Gemini Embeddings → Semantic Search → Top-K Context → Groq GPT-OSS 120B → Answer + Sources

4. Technology Stack

Component	Technology
UI / Application	Streamlit
PDF Text Extraction	PyMuPDF
Embeddings	Gemini gemini-embedding-2
Retrieval	Cosine Similarity / Top-K Semantic Search
LLM / Answer Generation	Groq — openai/gpt-oss-120b
Storage (MVP)	JSON files: chunks.json + embeddings.json
Secrets	.env locally / Streamlit Secrets for deployment

5. Legal Knowledge Base — 4 Documents

The current MVP uses these four Pakistani legal documents as its knowledge base:

#	Document
1	Constitution of the Islamic Republic of Pakistan 1973
2	Code of Criminal Procedure, 1898
3	Pakistan Penal Code, 1860
4	Muslim Family Laws Ordinance

The assistant should answer from these documents and clearly indicate when the available documents do not contain enough information.

6. Main Features

- Natural-language legal questions
- Semantic retrieval of relevant legal chunks

- Top-K context passed to the LLM
- Answer with source document, page and/or chunk reference
- Chat history using Streamlit session state
- Safe no-answer response when evidence is insufficient
- Legal-information disclaimer

7. Safety Rules

- Do not invent laws, sections, articles, punishments or legal requirements.
- Answer only from retrieved legal context.
- If relevant evidence is not found, clearly say that the available documents do not provide sufficient information.
- Present the system as an informational/educational assistant, not a replacement for a qualified lawyer.

8. Current Development Status

- Streamlit application selected for the MVP.
- Constitution PDF processed into legal chunks.
- Gemini embedding integration implemented.
- Groq generation model selected: openai/gpt-oss-120b.
- Four-document knowledge base defined.
- Remaining work: complete embeddings, add/reprocess all four documents, implement retrieval + RAG, connect the Streamlit UI, show sources, and test legal questions.

9. 2-Minute Hackathon Demo

1. Open the Streamlit app.
2. Ask: "Section 144 kya hai?"
3. Show the retrieved legal chunks.
4. Show the AI-generated answer and source reference.
5. Ask another question related to criminal law, constitutional law, or family law.
6. Ask an unrelated question and demonstrate the safe "information not found / insufficient evidence" behavior.

10. Success Criteria

The MVP is successful when it can answer common questions from the four uploaded Pakistani legal documents, retrieve relevant legal passages, show reliable sources, and avoid unsupported/hallucinated legal claims.